

Path 1: Pre Security > Module 4: Computer Fundamentals > Room 2: Computer Types

<https://tryhackme.com/room/computertypes>

Computer Types:

Explore the different types of computers, from laptops to the tiny chips inside your coffee machine.

>> Task 1: Introduction

Sophia was trying to connect a new device to her home's WiFi when she noticed something unexpected: "NexusCool Fridge X17." She laughed and wondered if connecting to the NexusCool Fridge X17 WiFi would let her download a cold meal. Setting the joke aside, she soon realized her neighbor had bought a smart refrigerator, a fridge with a built-in computer capable of connecting to the internet. It made her pause. Computers were no longer just laptops and phones. They were quietly infiltrating everyday objects, from kitchen appliances to doorbells. Sophia had seen strange stories online about smart devices behaving badly, but she had never really stopped to think about what these machines actually were.

♦ Learning Objectives:

Upon completion of this room, you will be able to identify and distinguish between different types of computers you use directly, such as laptops and smartphones, and indirectly, such as servers, IoT devices, and embedded systems. You will also understand what makes each type suited to its purpose.

>> Task 2: Sophia's Summer of Hidden Computer - Month 1

♦ The Computers You Sit in Front Of:

During her first month at Nova Labs, Sophia learned an important lesson.

Sophia learned that...

1. Not all computers are meant to move.
2. Not all computers are meant for people to sit in front of.

She was introduced to four types of computers that often look similar but serve very different purposes.

- Portable everyday computing
- Sustained performance at a fixed location

Let's compare these below:

Computer Type	Screen and Keyboard	Main Purpose
Laptop	Yes	Portable everyday computing.
Desktop	Yes	Sustained performance at a fixed location.
Workstation	Yes	Precision and reliability for professional tasks.
Server	No	Providing services to many users over a network.

♦ How Sophia Learned the Difference:

Sophia began with a **laptop**, the computer she already knew best. It was perfect for emails and documents, but when she pushed it with long tasks, it slowed down. Gabriel explained that laptops are built to be portable, and staying cool in a small, battery-powered device is difficult.

Next, she tried a **desktop**. It stayed in one place, used wall power, and had better cooling. The same task ran smoothly for much longer. Desktops are designed for consistency rather than mobility.

When Sophia moved on to professional work, such as simulations and 3D models, Gabriel showed her a **workstation**. It looked like a desktop, but it was built differently. Workstations prioritize accuracy and reliability, using specialized components to reduce errors during long or complex computations.

Finally, Gabriel took her to a room full of machines with no screens at all. These were **servers**. They ran continuously, answering requests from multiple users simultaneously. Sophia never touched them directly, but they powered the tools she used every day.

Q1) Which computer type usually runs without a dedicated screen and keyboard?

Answer: Server

Q2) What kind of computer with specialized components would one buy to carry out precision work?

Answer: Workstation

>> Task 3: Sophia's Summer of Hidden Computer - Month 2

By her second month at Nova Labs, Sophia had started noticing computers she had never interacted with directly. The most powerful computer most people own fits in their pocket, but millions more hide inside everyday objects: doors, lamps, coffee machines.

We continue to find computers hiding in everyday objects:

Type	What it is	Examples
Smartphone	Pocket-sized computer optimized for battery life and connectivity	iPhone, Android phone
Tablet	Touch-first computer with larger screen	iPad, drawing tablet
<u>IoT</u> device	Network-connected device with a single purpose	Thermostat, smart doorbell, fitness tracker
Embedded computer	Computer built into another device	Coffee maker controller, automatic door sensor, lamp dimmer chip

IoT vs Embedded: Both can be small and single-purpose. The difference is connectivity. IoT devices connect to a network to report data or receive commands. Embedded computers might not connect to anything; they do their job inside the machine, often for years without anyone knowing they exist.

Sophia walked through automatic doors every day at Nova Labs. She never realized that a tiny computer inside the door frame was detecting her movement and signaling the motor to open. That's embedded computing; invisible, reliable, everywhere.

Q1) What is the currently most popular pocket-sized computer?

Answer: Smartphone

Q2) What kind of computer would you expect to find in a coffee machine?

Answer: Embedded Computer

>> Task 4: Why Computers Come in Different Flavors

Sophia asked the question that had been bothering her all summer:

Sophia asked...

Why not just build one computer that does everything?

“Because every design is a trade-off,” replied Gabriel.

Mobility costs power. Smaller, portable computers must sacrifice sustained performance. Reliability costs money. Servers and critical systems use redundancy, such as extra power supplies and disks, to avoid failure.

Purpose shapes everything. You touch a phone. You ask a server for information. An IoT device works quietly without demanding attention.

There is no best computer. There is only the right tool for the job!